Quentin D. Read

Stats and data science to sustainably feed the world

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Professional appointments

Agricultural Research Service, U.S. Department of Agriculture, Raleigh, NC 2021–
Applied consulting statistician, Plant Science Research Office
- Support USDA researchers by designing experiments, processing and visualizing data, and doing statistical analyses using R, Stan, SAS, and Python
- Design and teach statistics and data science lessons and workshops to USDA researchers
- Assist researchers with git for collaborations and high-performance computing using Linux
- Pursue a research program modeling the impacts of the food system on human and natural communities, using techniques from ecology, environmental science, and economics
- Received 2024 Southeast Area award for customer service and technical expertise

National Socio-Environmental Synthesis Center (SESYNC), Annapolis, MD 2018-2021
Data scientist (2019-2021); *Postdoctoral fellow* (2018-2020)
- Provided data science consulting for socio-environmental research teams
- Maintained and wrote content for SESYNC's cyberhelp website using Markdown and Jekyll
- Developed and taught lessons for data science curriculum
- Modeled impacts of food waste and benefits of food waste reduction interventions
- Obtained $130,000 NSF fellowship for SESYNC's postdoctoral immersion training program
- Research featured in multiple media outlets (list)

Michigan State University (MSU), East Lansing, MI 2016-2018
Postdoctoral researcher, Department of Forestry
- Compiled, analyzed, and processed environmental and biodiversity datasets in R
- Designed spatial Bayesian models; did GIS analysis in R and GDAL

Education

Ph.D., Ecology & Evolutionary Biology, University of Tennessee, Knoxville, TN 2016
B. S., Environmental Science, University of North Carolina, Chapel Hill, NC 2009
Certified Carpentries data science instructor 2025

Publications (for full list see Google Scholar)

- 10 first-authored publications in journals including *PNAS*, *Resources Conservation & Recycling*, *Science of the Total Environment, Ecography,* and *Biology Letters*
- >120 co-authored publications in journals including *Science, Ecology, Science of the Total Environment,* and *Global Ecology and Biogeography*

Selected invited talks

- N.C. State University, Plant & Microbial Biology department seminar, Raleigh, NC 2022
- Duke University, University Program in Ecology Seminar Series, Durham, NC 2020
- Commission for Environmental Cooperation, Arlington, VA 2018

Selected software

- Lead developer, *Ostats*: R package for trait analysis of ecological communities 2021
- Co-developer, *ggalluvial*: R package adding functionality to ggplot2 2020
- Co-developer, *rslurm*: R package for running R code in parallel 2019

Selected teaching and mentoring experience

- Mentored student team that won grand prize in a 75-team data science competition 2021
- Designed and taught workshops on R, omics, mixed models, geospatial data, and git 2020-2025
- Taught undergraduate and graduate courses in biology and ecology 2012-2017
- Mentored 17 undergraduates; two published first-author research papers 2012-2020